Midpoint Lenses in Segment Spaces: Scale-Optimized Atomic Budgets and Averaged Endpoints

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Abstract

We study the actual real segment norms and represented atomic laws in *Midpoint lenses in segment spaces*. Retaining the coefficient energy and the squared head-test evaluations yields a convex, one-parameter replacement budget. Its optimizer strictly improves the source's retained two-square-root estimate whenever the moment data are nondegenerate; a two-node endpoint law realizes this improvement below the trivial radius bound. We prove a residual-aware version with shared-parameter interval control and give an exact derivative-sign certificate. Norm-weighted endpoint laws also transfer the sharp joint budget directly to an averaged quadratic lens estimate, without the usual factor-two loss. The same averaged-modulus constant is already announced for related tree-potential models; no historical priority is asserted. Infinite-space statements retain the source's coordinate-completion and finite-hull foundations. The nonrenorming and graph-distortion conclusions are not certified by finite moment computations. Contribution assessment: C3/5, a source-specific optimized atomic budget with realizable strict gain, residual control, and a direct weighted-endpoint transfer, not a new global renorming theorem.

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1 Scope, source architecture, and current context

The assigned September 27, 2026 manuscript [1] contains a reflexive forest-dual construction and an infinite-height coordinate predual, followed by several different proofs and quantitative versions of midpoint control. Its main estimate is

$$x = P_H x, \quad \|x+y\|, \|x-y\| \le R \quad \Longrightarrow \quad \|Q_H y\| \le 2\sqrt{R^2 - \|x\|^2}. \tag{1}$$

Here H is finite and ancestral and $Q_H = I - P_H$. In particular y need not have zero head. The source proves this by a joint coefficient and head-slack budget, not by treating the segment space as a Hilbert space. Its exact atomic decompositions, root gates, and opposite-sign comparison must be retained.

Source modules	Logical role and treatment here
Forest and infinite foundations; atomic; presentations	Define distinct completions, contractive projections, finite convex hulls and coordinate density. These are the foundations used below; the infinite predual is not the full dual.
Opposite crossing; joint budget; midpoint growth	Sharp source mechanism rederived with an arbitrary replacement scale in Sections 3–4.
Endpoint laws; stability; joint consequences	Weighted probability laws and residual means. Combined with the joint budget, rather than credited as newly invented tools.
Clipping; primal additions; radial endpoints	Alternative primal tests and direct averaged estimates, including the source coefficient 1/12. No clipping proof is replaced by a finite Gram computation.
Toolkit; coefficients; original coefficients; cancellation comparisons; sign trimming; target-tail signs	Alternative coefficient and sign budgets. Some deliberately lose constants or yield qualitative limits; they are not source errors.
Sequential cancellation; qualitative midpoint; obstruction; foundation variants	Passage through fixed heads and cofinite subspaces, and non-AUC renorming obstructions. The global obstruction is not independently reproved here.

Asymptotic midpoint uniform convexity (AMUC) and its lens formulation are classical in this context [2]. Baudier's September 18, 2026 preprint [3] proves AMUC for the Kadets–Werner modification of Bourgain–Rosenthal space and states its failure of AUC renormability. We inspected its main theorem, corollary, and short argument, not the proofs of its historical inputs. That example is different from the real segment models here. Perreau's work [4] concerns dual and weak-star structures, not an automatic identification of the two completions below.

The publisher's companion [6], whose abstract was freshly checked, already announces the averaged bound $\widehat{\delta}(t) \ge \sqrt{1+t^2/4} - 1$ for related bounded-tree-potential spaces. We therefore claim neither a new best universal constant nor priority for that expression. Section 5 supplies a direct route through this assigned manuscript's represented laws. The companion diamond manuscript [7] requires additional graph-selection and energy arguments. They are not reproduced or counted as contributions of this report.

2 The actual spaces and finite representations

All scalars below are real. A segment is a nonempty ancestral interval $[a, b]$ in a rooted tree. On finitely supported functions put

$$\rho(u) = \sup_{\substack{S_1,\dots,S_k \\ \text{pairwise disjoint segments}}} \left(\sum_{i=1}^{k} u(S_i)^2 \right)^{1/2}, \qquad u(S) = \sum_{v \in S} u(v). \tag{2}$$

The forest is the disjoint union of all finite-height, countably branching rooted trees, with heights unbounded. Let J be the completion in (2), and $X = J^*$. On one height-h component, $\|u\|_2 \le \rho(u) \le \sqrt{h+1}\,\|u\|_2$. The component decomposition is an outer ℓ_2 sum; the source uses this to prove reflexivity and coordinate density for X. Constants depending on h in this comparison cannot be used uniformly in the whole forest.

On $T_\infty = \mathbb{N}^{<\omega}$ use the same primal norm and completion J_∞, but take

$$B_\infty = \overline{\operatorname{span}}\{e_v^* : v \in T_\infty\} \ \subset J_\infty^*. \tag{3}$$

This is not all of J_∞^*. A functional summing along one infinite branch is a source example of a norm-one element at distance one from the coordinate predual. We write Y for either X or B_∞, with its stated norm.

A finite set H is ancestral if it contains the ancestors of each of its vertices in the same component. Restriction P_H and $Q_H = I - P_H$ are contractive on the relevant primal and dual spaces. Finite ancestral exhaustions converge in norm on Y. A finite vector supported in H has a norming primal test $p = P_H p$, $\rho(p) = 1$. These statements are source foundations; they are not implied by our finite examples.

For a segment S write $d_S(u) = u(S)$. A represented atom is the data

$$Z = \sum_i a_i d_{S_i}, \qquad S_i \text{ pairwise disjoint}, \qquad \sum_i a_i^2 \le R^2. \tag{4}$$

Its norm is at most R, by (2) and Cauchy–Schwarz. On a fixed finite ancestral set the radius-R ball is exactly the convex hull of these atoms. Indeed their support function at u is $R\rho(u)$: optimize the Euclidean coefficient vector for each disjoint family, and then take the finite maximum. Finite-dimensional separation identifies the ball. Caratheodory gives at most $|H| + 1$ atoms for a vector supported there. This proves existence, not an efficient decomposition algorithm. In the completed space one has norm approximation by finite laws, not an exact finite law for every vector.

Minimal vertices outside H whose ancestors lie in H, including roots of untouched components, are the gates $g \in G_H$. Their descendant cones U_g partition the complement. The source norm decomposes as

$$\|Q_H z\|^2 = \sum_g \left\| z|_{U_g} \right\|^2, \qquad \left(\sum_g |z(e_g)|^2 \right)^{1/2} \le \|Q_H z\|. \tag{5}$$

Here e_g is the primal coordinate vector. Intersecting a segment with a cone gives a segment. A segment crossing from H into U_g has tail $[g, w]$ and a head ending at the parent of g. In a disjoint family at most one segment uses each gate.

3 The gate calculation at an arbitrary scale

Let a finite probability space carry represented atoms $Z_\omega = \sum_i a_{\omega i} d_{S_{\omega i}}$, and let $|\sigma_\omega| \le 1$. The multiplier can depend on the whole representation. Put

$$x = \mathbb{E}Z, \quad y = \mathbb{E}(\sigma Z), \quad A = \mathbb{E}\sum_i a_i^2, \quad e = \mathbb{E}\sum_i s_i^2, \quad r = \mathbb{E}\sum_i a_i s_i, \quad s_i = p(S_i), \tag{6}$$

where $p = P_H p$ and $\rho(p) \le 1$. Thus $0 \le e \le 1$ and $r^2 \le Ae$. We assume $r \ge 0$, reversing p if needed. The quantities A, e depend on the represented law, not only on x, y. For $t \ge 0$ replace a_i by $b_i = ts_i$ and define

$$\begin{aligned} Z_t &= \sum_i b_i d_{S_i}, & V_t &= \sum_i (a_i - b_i)^2, & L_t &= t^2 - \sum_i b_i^2, \\ \eta_t &= \mathbb{E}V_t = A - 2rt + et^2, & \theta_t &= \mathbb{E}L_t = (1-e)t^2. \end{aligned} \tag{7}$$

Every quantity called a squared error or slack is nonnegative. Wholly exterior segments have $s_i = b_i = 0$; original exterior coefficients need not vanish.

Lemma 3.1 (Opposite crossings, retaining the source geometry)**.** *For two represented replacements and their crossing coefficients b_g, b'_g,*

$$2 \sum_{g : b_g b'_g < 0} |b_g b'_g| \le L_t + L'_t. \tag{8}$$

Proof. At a shared gate the two heads are nested. If the shorter head has test sum u and the longer one v, and $uv < 0$, splitting the longer head immediately before the shorter produces the squared-sum gain $(v-u)^2 + u^2 - v^2 = 2u^2 - 2uv \ge 2|uv|$. Split the corresponding full segment there. If the two heads are identical opposite signs cannot occur, since both coefficients are t times that same test sum. Charge the gain to the representation with the longer head. Different gates use different disjoint segments in each representation; hence all its charged splits can be performed simultaneously. The resulting disjoint segment family has squared test sum at most one. Multiplying by t^2 bounds the total gain charged to that representation by L_t, and similarly by L'_t for the other. When $t = 0$ both sides vanish. □

Theorem 3.2 (Scale-dependent and residual-aware lens budget)**.** *Under* (6)*, let $\kappa = (\sum_g |x(e_g)|^2)^{1/2}$. For every $t \ge 0$,*

$$\|Q_H y\| \le F_\kappa(A, r, e; t) := \sqrt{\eta_t} + \sqrt{(\kappa + \sqrt{\eta_t})^2 + 2\theta_t}. \tag{9}$$

In particular, if $x = P_H x$,

$$\|Q_H y\| \le f(t) := \sqrt{A - 2rt + et^2} + \sqrt{A - 2rt + (2-e)t^2}. \tag{10}$$

Independently, $\|Q_H y\| \le \sqrt{A}$, so the minimum of this bound and (9) *is valid.*

Proof. The representation of $Z - Z_t$ on disjoint segments gives $\|Z - Z_t\| \le \sqrt{V_t}$. Jensen and contractivity imply

$$\|Q_H \mathbb{E}[\sigma(Z - Z_t)]\| \le \sqrt{\eta_t}. \tag{11}$$

At a gate $Z_t(e_g) = b_g$. The original $Z(e_g)$ is either a crossing coefficient, a wholly exterior coefficient, or zero. The disjointness observation therefore gives $\sum_g |Z(e_g) - b_g|^2 \le V_t$, and $\|\mathbb{E}b\|_{\ell_2} \le \kappa + \sqrt{\eta_t}$. For independent copies b, b',

$$\begin{aligned} \|\mathbb{E}|b|\|_{\ell_2}^2 - \|\mathbb{E}b\|_{\ell_2}^2 &= \mathbb{E}\sum_g (|b_g b'_g| - b_g b'_g) \\ &= 2\mathbb{E}\sum_{b_g b'_g < 0} |b_g b'_g| \le 2\theta_t. \end{aligned} \tag{12}$$

Independence is used only in this product-of-means identity, not between σ and Z. Each cone contribution of $Q_H Z_t$ is $b_g d_{[g,w]}$, of norm $|b_g|$. The triangle inequality in that cone and (5) yield $\|Q_H \mathbb{E}(\sigma Z_t)\| \le \|\mathbb{E}|b|\|_{\ell_2}$. Combine this with (11) and (12). Finally $\|Q_H \mathbb{E}(\sigma Z)\| \le \mathbb{E}\,\|Z\| \le \sqrt{A}$. □

At the source scale $t = r$ with zero mean tail, the retained estimate is

$$f(r) = \sqrt{A - (2-e)r^2} + \sqrt{A - er^2} \le 2\sqrt{A - r^2}. \tag{13}$$

The order of the two radicands follows from (7); interchanging them in a symmetric sum changes nothing, but their derivatives must follow their actual definitions. The source finally uses $A \le R^2$. Our next section keeps A, e and changes t.

4 Certified scalar optimization

Proposition 4.1 (Strict gain over the retained source bound)**.** *Suppose $r > 0$, $0 < e < 1$, and $Ae > r^2$. Then f in* (10) *is strictly convex on $[0, \infty)$ and has a unique minimizer t_* satisfying*

$$r < t_* < r/e, \qquad f(t_*) < f(r). \tag{14}$$

Proof. For $q_c(t) = A - 2rt + ct^2$,

$$(\sqrt{q_c})' = \frac{ct - r}{\sqrt{q_c}}, \qquad (\sqrt{q_c})'' = \frac{cA - r^2}{q_c^{3/2}}. \tag{15}$$

Both $c = e$ and $c = 2 - e$ give positive radicands and positive second derivatives. Moreover $f'(0) = -2r/\sqrt{A} < 0$ and

$$f'(r) = (1-e)r\left(\frac{1}{\sqrt{A - er^2}} - \frac{1}{\sqrt{A - (2-e)r^2}}\right) < 0. \tag{16}$$

At $t = r/e$ the first derivative term is zero and the second is positive. Strict convexity and continuity prove the claim. □

This is an optimization of a proved upper envelope, not of the actual segment norm or the universal lens constant. Its strict gain can be arbitrarily small when e tends to one. If $r = 0$, $t = 0$ minimizes. If $e = 0$, Cauchy forces $r = 0$. If $e = 1$, $t = r$ minimizes and gives $2\sqrt{A - r^2}$. When $Ae = r^2$, the first square root can have a cusp; convex minimization remains valid but the smooth proof does not.

Proposition 4.2 (Exact sign test and a finite optimization error)**.** *In the strict case of Proposition 4.1, for $r \le t \le r/e$ the sign of $f'(t)$ equals the sign of*

$$P(t) = ((2-e)t - r)^2 q_e(t) - (r - et)^2 q_{2-e}(t). \tag{17}$$

Thus rational data admit rational derivative bracketing without squaring an unknown sign. If $l \le t_* \le u$ is such a bracket and Lipschitz constant $L_f = \sqrt{e} + \sqrt{2-e}$ is used, then at its midpoint v,

$$f(v) - L_f(u-l)/2 \le f(t_*) \le f(v). \tag{18}$$

Proof. On this interval $(2-e)t - r \ge 0$ and $r - et \ge 0$. Comparing the two nonnegative derivative terms after multiplication by positive denominators gives (17); only then is squaring valid. Each square root is the Euclidean norm of $(\sqrt{c}(t - r/c), \sqrt{A - r^2/c})$, so it is $\sqrt{c}$-Lipschitz. Add and use $|v - t_*| \le (u-l)/2$. A trial value already supplies an upper bound; locating the optimizer exactly is unnecessary. □

Example 4.3 (A realizable two-node law)**.** Take a root o, one child g, $H = \{o\}$ and $p = e_o$. The plus endpoint law chooses $d_{\{o\}}$ with probability $4/5$ and $d_{[o,g]}$ with probability $1/5$. The minus endpoint law chooses $d_{\{o\}}$ with probability $4/5$ and $-d_{\{g\}}$ with probability $1/5$. Mix the laws equally with $\sigma = +1, -1$. Every represented atom has squared coefficient energy one. Direct calculation gives

$$x = \tfrac{9}{10} e_o^*, \quad y = \tfrac{1}{10} e_o^* + \tfrac{1}{5} e_g^*, \quad A = 1, \quad r = e = \tfrac{9}{10}, \quad \|Q_H y\| = \tfrac{1}{5}. \tag{19}$$

At the rational trial scale $t = 37/40$,

$$\begin{aligned} f(9/10) &= \sqrt{109/1000} + \sqrt{271/1000}, \\ f(37/40) &= \sqrt{1681/16000} + \sqrt{4419/16000}. \end{aligned} \tag{20}$$

Exact rational square-root enclosures give $0.8507 < f(9/10) < 0.8508$ and $0.8496 < f(37/40) < 0.8498$. Thus the gain is strict and both estimates are below the trivial radius one. The actual tail is $1/5$; these estimates are not advertised as sharp. Representation metadata, including the singleton in the minus law, is essential to $e = 9/10$.

Example 4.4 (A cusp diagnostic, not a realizability assertion)**.** For $A = 1$, $e = r^2$, $0 < r < 1$, elementary convex minimization gives

$$(t_*, f(t_*)) = \begin{cases} (2r/(2-r^2),\ 4(1-r^2)/(2-r^2)), & r^2 \le 2/3, \\ (1/r,\ \sqrt{2(1-r^2)/r^2}), & r^2 \ge 2/3. \end{cases} \tag{21}$$

Indeed $f(t) = |1 - rt| + \sqrt{1 - 2rt + (2-r^2)t^2}$. To the left of $1/r$ its derivative vanishes at $2r/(2-r^2)$, which lies there exactly in the first regime. To the right both derivative contributions are positive. Otherwise the cusp minimizes. At $r = 9/10$ the value is $\sqrt{38}/9$. Feasible moment triples do not by themselves construct segment endpoint laws; this example tests the scalar boundary only.

5 Averaged endpoints without a factor-two conversion

The preceding argument requires the pooled mean to be in the head. It does not require the plus and minus outcomes to have probability one half. This permits a useful combination of the source's weighted endpoint laws with its sharp joint budget.

Theorem 5.1 (Norm-weighted averaged lens)**.** *For either $Y = X$ or $Y = B_\infty$, every finite ancestral H, every $x = P_H x$ and every $y \in Y$, put $L = (\|x+y\| + \|x-y\|)/2$. Then*

$$\|Q_H y\| \le 2\sqrt{L^2 - \|x\|^2}, \qquad L \ge \sqrt{\|x\|^2 + \|Q_H y\|^2}/4. \tag{22}$$

Proof. First let x, y be supported in a finite ancestral set $K \supseteq H$. Write $a = \|x+y\|$ and $b = \|x-y\|$. If $a = 0$ or $b = 0$ then $y = \mp x$, hence $Q_H y = 0$, and the triangle inequality suffices. Otherwise choose represented unit-atom laws with respective means $(x+y)/a$ and $(x-y)/b$. Multiply every atom by L and mix the two laws with probabilities $a/(2L)$ and $b/(2L)$. Assign multipliers $+1, -1$ respectively. The resulting represented law satisfies

$$\mathbb{E}Z = x, \qquad \mathbb{E}(\sigma Z) = y, \qquad \sum_i a_i^2 \le L^2 \text{ in every outcome.} \tag{23}$$

Choose a head norming test with $r = x(p) = \|x\|$; for $x = 0$ use $p = 0$. Theorem 3.2 at $t = r$ and (13) give $\|Q_H y\| \le 2\sqrt{A - r^2} \le 2\sqrt{L^2 - r^2}$.

For general y, choose finite ancestral $K_j \supseteq H$ exhausting its coordinates, and set $y_j = P_{K_j} y$. The source's completion theorem gives $y_j \to y$ in norm. The preceding estimate applies to $L_j = (\|x+y_j\| + \|x-y_j\|)/2$. Contractivity gives $L_j \le L$, while norm convergence gives $L_j \to L$ and $Q_H y_j \to Q_H y$. Passage to the limit proves (22). There is no interchange of a uniform law over all finite heads with an unproved infinite atomic representation. □

Corollary 5.2 (Averaged asymptotic modulus)**.** *For these source spaces define*

$$\widehat{\delta}_Y(u) = \inf_{\|v\|=1} \sup_{\substack{F \subseteq Y \text{ closed} \\ \operatorname{codim} F < \infty}} \inf_{\substack{w \in F \\ \|w\| \ge 1}} \left(\frac{\|v + uw\| + \|v - uw\|}{2} - 1 \right). \tag{24}$$

Then, for every $u > 0$,

$$\widehat{\delta}_Y(u) \ge \sqrt{1 + u^2/4} - 1. \tag{25}$$

Proof. Fix a unit v and $\varepsilon > 0$. Choose finite ancestral H with $x = P_H v$ and $\|v - x\| < \varepsilon$, so $\|x\| \ge 1 - \varepsilon$. Choose the closed finite-codimensional subspace $F = \ker P_H$ before choosing w. For every $w \in F$ of norm at least one, Theorem 5.1 and endpoint norm Lipschitz continuity give

$$\frac{\|v + uw\| + \|v - uw\|}{2} \ge \sqrt{(1-\varepsilon)^2 + u^2/4} - \varepsilon$$

for $0 < \varepsilon < 1$. Take the infimum over w, the supremum over F, and then let $\varepsilon \downarrow 0$. The bound is independent of v, so the outer infimum preserves it. □

For the source's usual range $0 < u < 1$, this improves its bound $\frac{1}{2}(\sqrt{1 + u^2/4} - 1)$ obtained by converting its maximum modulus, and also its direct clipping bound $\sqrt{1 + u^2/12} - 1$. It is a combination of valid source ingredients, not a correction of either source proof. The identical expression in [6] precludes a claim of first discovery for the constant; the present proof uses these particular segment laws. It does not prove that the coefficient $1/4$ is optimal.

In Example 4.3, the endpoints are $e_o^* + \frac{1}{5} e_g^*$ and $\frac{4}{5} e_o^* - \frac{1}{5} e_g^*$. The first has norm one: it is a convex combination of $d_{\{o\}}$ and $d_{[o,g]}$, and evaluates to one at e_o. For the second, the Euclidean upper bound from the two singleton segments is $\sqrt{17}/5$. The primal vector $(4, -1)/\sqrt{17}$ has singleton squared sum one and segment sum $3/\sqrt{17} < 1$, so it attains that bound. Hence

$$L = \frac{5 + \sqrt{17}}{10}, \qquad 2\sqrt{L^2 - (9/10)^2} = \frac{\sqrt{10\sqrt{17} - 39}}{5} < \frac{3}{10}.$$

The last inequality follows by squaring positive quantities, since $\sqrt{17} < 33/8$. This bound uses the actual endpoint norms and is much better than the scale optimization of the original radius-one law in Example 4.3. Different certified laws and different budgets should be compared, not automatically combined as if their moments belonged to the same representation.

6 Moment enclosures and physical residuals

The Ram comparison is a finite representation with its actual pairings and omitted terms retained. It is not a claim that (2) is generated by a single positive Gram matrix. Original TR250 [8] requires a physical space before moment coordinates and proves a residual account with source and boundary defects. The original physical-metric chapter [9] likewise separates a compressed residual from its unobserved component. Their Hilbert Pythagoras identities cannot be copied unchanged into this Banach norm. Here the correct replacement is the disjoint-segment estimate and the cone norm (5), both proved for the actual source geometry. The later comparison TR-item-047 [10] is used only as a residual-aware finite-certificate comparison, not as an original proof of a segment-space theorem.

Proposition 6.1 (Shared-parameter interval certificate)**.** *Suppose an actual represented law satisfies* (6)*, and certified numbers obey*

$$A \le A_+, \quad 0 \le r_- \le r, \quad e \le e_+ \le 1, \quad \kappa \le \kappa_+, \quad A_+ e_+ \ge r_-^2.$$

Let a target $\widetilde{y}$ obey $\|Q_H(\widetilde{y} - \mathbb{E}\sigma Z)\| \le \varepsilon$. For any $t \ge 0$, set

$$\eta_+ = A_+ - 2r_- t + e_+ t^2, \qquad \theta_+ = (1 - e_+)t^2.$$

Then

$$\|Q_H \widetilde{y}\| \le \varepsilon + \sqrt{\eta_+} + \sqrt{(\kappa_+ + \sqrt{\eta_+})^2 + 2\theta_+}. \tag{26}$$

In particular the same e_+ appears with opposite signs in the two linked terms; no independent maximization of incompatible slacks is needed.

Proof. The expression F_κ increases with A and κ and decreases with r for fixed $t \ge 0$. Increase A first, decrease r next, and then increase e, so all intermediate quadratic radicands remain admissible. For $\eta > 0$, differentiation with respect to e gives

$$\frac{\partial F_\kappa}{\partial e} = \frac{t^2}{2\sqrt{\eta}} \left(1 + \frac{\kappa - \sqrt{\eta}}{\sqrt{(\kappa + \sqrt{\eta})^2 + 2(1-e)t^2}} \right) \ge 0. \tag{27}$$

The denominator is at least $|\kappa - \sqrt{\eta}|$ because $\kappa, \eta, 1 - e$ are nonnegative. At a zero denominator or $\eta = 0$, use continuity from admissible positive radicands. Theorem 3.2 and the stated target residual complete the proof. If $e_+ = 0$, feasibility forces $r_- = 0$ and the same formula is directly meaningful. □

The proposition has concrete prerequisites. A finite law supplies the segments, their disjointness, the nonnegative probabilities, and the coefficient lists. All three moments in (6) can then be evaluated exactly for rational data. On finite support, $\rho(p) \le 1$ can be checked by enumerating the disjoint segment families; this is a finite test, not a practical complexity claim. A matrix of observed head values alone need not certify $\rho(p) \le 1$. Nor does the head moment error bound the target tail residual ε.

If the finite law approximates a completed vector, a genuine bound on its omitted tail in Y is necessary for (26). Gate means are weaker information than the whole tail norm, although their ℓ_2 norm is enough for the particular calculation (12). The source's coordinate convergence provides qualitative approximation; it supplies no automatic effective cutoff from a fixed number of observations.

7 What is preserved, and what remains global

The source has several valid mechanisms which are not displaced by the local optimization. Cumulative clipping constructs two inward primal tests with a controlled quadratic norm cost. Original coefficient cancellation tracks both the signs of coefficients and the squared head sums. Sign-trimming arguments control weighted discarded mass, not merely a small probability of bad atoms. The source's qualitative separate-normalization argument correctly keeps its nonzero pooled tail until it proves that tail tends to zero. Our exact norm-weighted mixture avoids that particular residual only when its stated endpoint norms and laws are used.

For nonrenormability the source uses bounded root-path vectors and weakly null normalized child increments in the full relevant dual topology. Under an alleged positive AUC modulus, repeated choices outside finite-codimensional spaces force multiplicative path growth, contradicting bounded path norms. Finite forest heights must be chosen large enough in advance, while the infinite-height argument uses a different completion. None of those quantifiers follows from the three numbers (A, r, e), and the argument is not claimed as a new theorem of this report.

Similarly, a finite example with a small optimized envelope does not prove a universal graph-distortion lower bound. M639's companion problem uses additional countably branching selection and tree energy inequalities. This report concerns the local lens input and its fixed-head passage to a modulus. No graph recurrence from that companion is counted here.

8 Advantages, costs, and further directions

Advantages. The optimized certificate retains the source's representation rather than replacing its norm by an unrelated Euclidean surrogate. Three scalar moments and an actual gate-mean residual determine a valid envelope. Strict improvement over the retained source scale is proved, and Example 4.3 shows that the gain is realizable and can survive the trivial radius cap. The averaged theorem works for arbitrary head displacement, not only a pure-tail y.

Costs and limitations. Exact finite hull representations may be expensive to obtain. Their moment data are noncanonical: a different convex decomposition can change A, e without changing the endpoints. A sharper scalar optimizer does not imply sharper underlying geometry. When $e = 1$ there is no scale gain, and near that boundary the gain can be tiny. Interval error can overwhelm it. No runtime comparison, optimal lens constant, new AUC separation, or whole-corpus proof audit is claimed.

Extensions requiring new work. One could optimize jointly over admissible atomic representations and t. That would require an exact description or certified relaxation of the finite convex-hull constraints, not only the necessary inequality $r^2 \le Ae$. One could also seek quantitative tail approximations in the completed spaces. Such a program needs additional information on the actual vector's coordinate tail. Finally, a complex-scalar analogue would need a phase-sensitive replacement for ordered opposite real signs; the present proof does not establish it.

9 Evidence, attribution, and contribution

The literal-read ledger records all 29 cached source text files, including the scientific modules, preamble, bibliography, and README, with hashes before the displays and matching hashes after them. Literal coverage is not an independent verification of every theorem or every cited external proof. The architecture table distinguishes the rederived gate mechanism from imported foundations and downstream global claims.

Fresh original-private reads comprise TR250 lines 121–181 and 455–599 (physical spaces, finite moments, the hypercircle proof and its residual qualifications), and the physical-metric chapter lines 1–210. The later comparison TR-item-047 was read only at lines 346–411. Included figures and their generating code were not read as proofs. All other private passages remain unread for this assignment; exact complements are in the ledger.

The owned executable checks test rational law moments, the signed derivative comparison, cusp formulas, and outward rational square-root enclosures: 4,572 exact cases and 4,711 assertions, with no floating-point diagnostic cases. Of these, 4,464 cases enumerate pairs of disjoint segment families on a fixed four-vertex tree with two gates. They supplement the proofs and do not test all infinite segment configurations. Parent checks are separate and excluded from the author counts. Final counts and hashes are recorded in the accompanying owned evidence receipts.

The principal increment is the adjustable replacement theorem, its strict improvement and realizable example, and the shared-parameter residual certificate. The weighted-endpoint derivation is a useful sharpening within the assigned source, with explicit acknowledgment of the companion constant. C3 is assigned for this combined, proved local increment. Applicability L3 remains a separate relevance classification. No error in the source's main lens proof is alleged, and no historical novelty claim is made for convexity, Cauchy–Schwarz, convex hulls, or one-dimensional optimization.

References

[1] OpenAI, *Midpoint lenses in segment spaces*, September 27, 2026, captured source release. Publisher source. All cached text modules read; source byte provenance is separate from proof verification.

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